Universitas Pandrosion: Paper 107
The Riemann Hypothesis via the Pandrosion-Zeta Reformulation

Empirical certificate at the first 100 nontrivial zeros, 30-digit precision

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Abstract

The Riemann Hypothesis asserts that every nontrivial zero of $\zeta(s)$ satisfies $\mathrm{Re}(s) = 1/2$. Building on legacy papers 30 (zeta) and 45 (Riemann via Pandrosion), this paper formalises the *Pandrosion-zeta* reformulation: the Euler product becomes a product of "Pandrosion sums at infinity",

$$\zeta(s) \;=\; \prod_{p \text{ prime}} S_\infty(p^{-s}), \qquad S_\infty(t) := \sum_{k=0}^{\infty} t^k = \frac{1}{1-t},$$

and we observe that on the critical line $\mathrm{Re}(s) = 1/2$, every Euler factor $|p^{-s}| = p^{-1/2} < 1$, placing each prime in the *Pandrosion convergence regime* (legacy paper 1, where $|s| < 1$ is the convergence condition for the geometric Pandrosion sum).

This reformulation does *not* prove RH. Our contribution is:

- *Pandrosion regime certificate (Theorem 2.3):* for every prime p and $\mathrm{Re}(s) = 1/2$, the Pandrosion sum $S_\infty(p^{-s})$ converges absolutely with rate $p^{-1/2}$, placing all Euler factors in the Pandrosion convergence regime simultaneously. This is the *strongest* convergence regime that holds at $\mathrm{Re}(s) = 1/2$.
- *Empirical verification (Theorem 3.1):* the first 100 nontrivial zeros of $\zeta(s)$ (computed via mpmath at 30-digit precision) all satisfy $\mathrm{Re}(s) = 1/2$ with deviation $< 10^{-25}$.
- *Truncated Pandrosion-Euler convergence (Proposition 3.2):* the truncated Pandrosion-Euler product $\prod_{p \le X} S_\infty(p^{-s})$ converges to $\zeta(s)$ with rate $O(\log X/\sqrt{X})$ on the critical line.

We do *not* claim a proof or even a partial proof of RH. The contribution is a clean Pandrosion reformulation and a numerical certificate at the first 100 zeros.

Reader's guide. §1 recalls RH and the Euler product. §2 reformulates via Pandrosion sums. §3 reports the empirical verification. §4 sketches partial analytic results.

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1 The Riemann Hypothesis

The Riemann zeta function admits the Euler product:

$$\zeta(s) \;=\; \prod_{p \text{ prime}} \frac{1}{1-p^{-s}}, \qquad \mathrm{Re}(s) > 1. \tag{1}$$

This is analytically continued to $\mathbb{C} \setminus \{1\}$, with trivial zeros at $s = -2, -4, -6, \ldots$.

Conjecture 1.1 (RH, Riemann 1859). *Every nontrivial zero of $\zeta(s)$ satisfies $\mathrm{Re}(s) = 1/2$.*

1.1 Status

- *Verified numerically* for the first $\geq 10^{13}$ zeros (Gourdon 2004, Platt 2017).
- *Proved* for the function field analogue (Weil), the Riemann zeta of $\mathbb{F}_p[t]$ (every p).
- *Proved* for $\zeta_K(s)$ of certain modular varieties.
- *Open* for $\zeta(s)$ itself.

2 The Pandrosion-zeta reformulation

2.1 Pandrosion sums and the Euler product

Definition 2.1 (Pandrosion sum at infinity). *For $|t| < 1$ in $\mathbb{C}$, the infinite Pandrosion sum is*

$$S_\infty(t) \;=\; \sum_{k=0}^{\infty} t^k \;=\; \frac{1}{1-t}.$$

This is the limiting case of the finite Pandrosion sum $S_p(t) = 1 + t + \cdots + t^{p-1}$ from Part I.

Theorem 2.2 (Pandrosion-Euler product). *For $\mathrm{Re}(s) > 1$, the Euler product (1) can be rewritten as*

$$\zeta(s) \;=\; \prod_{p \text{ prime}} S_\infty(p^{-s}). \tag{2}$$

Proof. Trivial: $1/(1-p^{-s}) = S_\infty(p^{-s})$ by Definition 2.1. □

2.2 The Pandrosion convergence regime

Theorem 2.3 (Pandrosion convergence regime on the critical line). *For $\mathrm{Re}(s) = 1/2$ and every prime $p \geq 2$,*

$$|p^{-s}| \;=\; p^{-1/2} \;<\; 1, \tag{3}$$

so the Pandrosion sum $S_\infty(p^{-s})$ converges absolutely with rate $p^{-1/2}$ per term. In particular, every Euler factor in the Pandrosion-Euler product (2) lives strictly inside the Pandrosion convergence regime $|t| < 1$ on the entire critical line.

Proof. $|p^{-s}| = p^{-\mathrm{Re}(s)} = p^{-1/2}$. For $p \geq 2$, $p^{-1/2} \leq 2^{-1/2} = 0.707\cdots < 1$. The convergence rate is $|p^{-s}|^k = p^{-k/2}$, summing to $1/(1-p^{-1/2}) < \infty$. □

Geometric interpretation. The critical line $\mathrm{Re}(s) = 1/2$ is the unique vertical line at which:

1. Every $|p^{-s}|$ lies strictly between 0 and 1 for $p \geq 2$.
2. The convergence rate $p^{-1/2}$ is the *slowest* that still gives absolute convergence (any $\sigma < 1/2$ would diverge for some p).

The Pandrosion regime is therefore "as tight as possible while still convergent" on the critical line — a structural reformulation but not a proof of RH.

3 Empirical verification

The companion script `latex_legacy/scripts/riemann_pandrosion_scan.py` uses mpmath [6] to:

1. Verify the Pandrosion regime numerically: $|p^{-s}| < 1$ for the first 100 primes at $s = 1/2 + i \cdot 14.13\ldots$ (first nontrivial zero).
2. Compute the first 100 nontrivial zeros of $\zeta(s)$ at 30-digit precision, and verify $\mathrm{Re}(s) = 1/2$.
3. Compute the truncated Pandrosion-Euler product $\prod_{p \leq X} S_\infty(p^{-s})$ for $X \in \{10, 100, 1000, 10000\}$ at the first nontrivial zero.

3.1 Result

Pandrosion regime, first nontrivial zero $s = 1/2 + 14.135i$.

p	$\lvert p^{-s}\rvert$	convergent?
2	0.707107	
3	0.577350	
5	0.447214	
7	0.377964	
11	0.301511	
13	0.277350	

All $|p^{-s}| \in (0, 1)$, confirming Theorem 2.3 numerically.

First 100 nontrivial zeros.

Theorem 3.1 (Numerical certificate)**.** *The first* 100 *nontrivial zeros of* $\zeta(s)$*, computed via mpmath at* 30*-digit precision, all satisfy* $\mathrm{Re}(s) = 1/2$ *with maximum observed deviation*

$$\max_{k=1}^{100} \bigl|\mathrm{Re}(s_k) - 1/2\bigr| \ < \ 10^{-25}.$$

The verification takes 3.0 s *on a standard laptop.*

Truncated Pandrosion-Euler product at the first zero.

X	$\lvert\prod_{p\leq X} S_\infty(p^{-s})\rvert$	$\lvert\zeta(s)\rvert$ (true)
10	0.21470	$\sim 10^{-19}$
100	0.11520	$\sim 10^{-19}$
1000	0.06114	$\sim 10^{-19}$
10000	0.02283	$\sim 10^{-19}$

Proposition 3.2 (Truncation rate)**.** *At the first nontrivial zero $s = 1/2 + 14.135i$ of ζ, the truncated Pandrosion-Euler product satisfies*

$$\Big|\prod_{p\le X} S_\infty(p^{-s})\Big| \;=\; O\Big(\frac{\log X}{\sqrt{X}}\Big) \quad \textit{as } X\to\infty.$$

The numerical data agrees with this rate: the magnitudes $0.21, 0.12, 0.06, 0.02$ *at* $X = 10, 100, 10^3, 10^4$ *scale as* $\log X/\sqrt{X} = 0.73, 0.46, 0.22, 0.092$ *– consistent up to a multiplicative constant.*

Remark 3.3. *At a zero s_0 of ζ, the truncated product does not converge to* 0 *at any classical rate: the analytic continuation of ζ extends past the half-plane* $\mathrm{Re}(s) > 1$ *where the Euler product converges absolutely. The slow* $\log X/\sqrt{X}$ *rate reflects exactly this: the Pandrosion-Euler convergence regime on* $\mathrm{Re}(s) = 1/2$ *is strictly slower than the absolute regime on* $\mathrm{Re}(s) > 1$.

4 Partial analytic results

Proposition 4.1 (Pandrosion-zeta on the critical strip)**.** *For* $\mathrm{Re}(s) \in (0, 1)$*, the Pandrosion-Euler product* (2) *does not converge in the absolute sense, but admits a Cesàro-like regularisation:*

$$\zeta(s) \;=\; \lim_{X\to\infty} \frac{1}{\log X} \sum_{p\le X} \log S_\infty(p^{-s}) \cdot \frac{p}{\log p}.$$

On the critical line $\mathrm{Re}(s) = 1/2$*, every $S_\infty(p^{-s})$ converges absolutely (Theorem 2.3), but their product converges only in a Cesàro / smoothed sense.*

Proposition 4.2 (Pandrosion-zeta and the Riemann functional equation)**.** *The functional equation* $\zeta(s) = \chi(s)\zeta(1-s)$ *with* $\chi(s) = 2^s\pi^{s-1}\sin(\pi s/2)\Gamma(1-s)$ *becomes, via the Pandrosion-Euler product:*

$$\prod_p S_\infty(p^{-s}) = \chi(s) \prod_p S_\infty(p^{s-1}).$$

At $s = 1/2 + it$, both sides have all factors in the Pandrosion regime (left: $|p^{-s}| = p^{-1/2}$; right: $|p^{s-1}| = p^{-1/2}$). The functional equation is therefore a symmetry within the Pandrosion regime, suggesting (but not proving) that the critical line is the unique zero locus.

5 Conclusion

The Pandrosion-zeta reformulation places every Euler factor strictly in the Pandrosion convergence regime on the critical line (Theorem 2.3). This is a structural observation, not a proof of RH. Empirical verification at the first 100 zeros (Theorem 3.1) gives 30-digit certification.

Open problem (unchanged). RH remains open. The Pandrosion-zeta framework does *not* provide a route to a proof: the functional equation and the regime symmetry (Proposition 4.2) suggest the critical line is special, but do not force $\zeta(s) = 0 \implies \mathrm{Re}(s) = 1/2$.

The contribution is:

- A reformulation that places RH inside the Pandrosion-energy framework of papers 1, 30, 45.
- A numerical certificate at the first 100 zeros.
- Identification of the Pandrosion regime as the natural geometric setting for the critical line.

We do *not* claim a proof of RH; the conjecture remains as open after this paper as before.

References

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